milk, semen, vaginal secretions, urine bile and feces of these semen and saliva are known to transmit the **infection** very commonly. Transfusion of carrier blood once the most widely known mode infection has largely been eliminated by donor screening is strictly enforced. Infection by direct contact with open skin lesions such as pyoderma, eczema, cuts and scratches is very common among **young children** in developing countries. Certain groups and **occupations** carry a high risk of infection. These include medical and paramedical staff of blood banks, dialysis units, barbers, box workers.

The incubation period is long about 1- 6 months. The onset is insidious and fever is not prominent. Extra hepatic complications like arthralgia, urticaria and glomerulonephritis may occur. About 90-95% of adults with acute hepatitis B infection recover within 1-2months of onset and eliminate the virus from the body. They may be **Asymptomatic carriers** or may progress to recurrent or chronic liver disease.

Laboratory Diagnosis

Serology

Diagnosis of hepatitis B depends on the **serological demonstration** of the viral markers. **HBsAg** is the first marker to appear in **blood** after infection, being detectable. It remains in circulation throughout the symptomatic course of the disease (2- 6months). Anti HBs is the protective antibody.

HBcAg is **not demonstrable** in circulation because it's **enclosed** within the **HBsAg coat** but its antibody, anti HBc appears in serum a week or two after the appearance of HBsAg. As anti HBc remains life long, it serves as a useful indicator of prior infection with HBv.

HBeAg appears in blood concurrently with HBsAg, indicating the **high infectivity**. Molecular methods such as **DNA: DNA hybridization and PCR** at present used for HBV DNA testing are highly sensitive and quantitative.

Immunization

Both passive and active methods of immunization are available. **Active immunization** is more effective. The currently preferred vaccine is genetically engineered by cloning the **S gene for HBV** in **Baker's yeast**. A special vaccine containing all antigenic components of HBsAg (Pre-S1, Pre-S2 and s) has been developed. **No specific antiviral treatment** is available for acute HBV infection.

10.5 Rabies Virus

The Family Rhabdoviridae contains viruses that **infects mammals, reptiles, birds, fishes, insects and plants**. The disease in human being is called **hydrophobia** because the patient exhibits **fear of water**, being **incapable** of drinking though subject to **intolerable thirst**.

Pasteur established that the rabies virus was present in the **brain of infected animals**. By serial **intracerebral passage** in rabbits, he demonstrated **fixed virus** that could be **rendered immune** by a series of injections. **Vaccine was prepared** by drying pieces of **spinal card** from rabbits infected with the **fixed virus**.

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